List of Publications

Alberto De Marchi

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Papers listed by type and year. List available online in BibTEX format here.bib.
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Journal articles

[Ber+26] Jeremy Bertoncini, Alberto De Marchi, Matthias Gerdts, and Simon Gottschalk. "Reinforcement learning for adaptive interior point methods in convex quadratic programming". In: *Computational Optimization and Applications* (2026). in press. arXiv: 2509.07404.

[DHM26] Alberto De Marchi, Tim Hoheisel, and Patrick Mehlitz. "Augmented Lagrangian methods for fully convex composite optimization". In: *Mathematics of Operations Research* (2026). in press. arXiv: 2511.07117.

[Gol+26] Maxence Gollier, Mohamed Laghdaf Habiboullah, Geoffroy Leconte, Robert Baraldi, Alberto De Marchi, Dominique Orban, and Youssef Diouane. "RegularizedOptimization.jl: A Julia framework for regularized and nonsmooth optimization". In: *Journal of Open Source Software* 11.118 (2026), p. 9344. DOI: 10.21105/joss.09344.

[De 25] Alberto De Marchi. "Mixed-integer linearity in nonlinear optimization: a trust region approach". In: *Optimization Letters* 19.9 (2025), pp. 1883–1904. DOI: 10.1007/s11590-025-02190-9. arXiv: 2310.17285.

[DT25] Alberto De Marchi and Andreas Themelis. "A penalty barrier framework for nonconvex constrained optimization". In: *Journal of Nonsmooth Analysis and Optimization* 5.14585 (2025). DOI: 10.46298/jnsao-2025-14585. arXiv: 2406.09901.

[NDG25] Viktoriya Nikitina, Alberto De Marchi, and Matthias Gerdts. "Hybrid optimal control with mixed-integer Lagrangian methods". In: *IFAC-PapersOnLine* 59.19 (2025). 13th IFAC Symposium on Nonlinear Control Systems NOLCOS 2025, pp. 585–590. DOI: 10.1016/j.ifacol.2025.11.098. arXiv: 2403.06842.

[De 24b] Alberto De Marchi. "Implicit augmented Lagrangian and generalized optimization". In: *Journal of Applied and Numerical Optimization* 6.2 (2024), pp. 291–320. DOI: 10.23952/jano.6.2024.2.08. arXiv: 2302.00363.

[DM24] Alberto De Marchi and Patrick Mehlitz. "Local properties and augmented Lagrangians in fully nonconvex composite optimization". In: *Journal of Nonsmooth Analysis and Optimization* 5 (2024). DOI: 10.46298/jnsao-2024-12235. arXiv: 2309.01980.

[DT24] Alberto De Marchi and Andreas Themelis. "An interior proximal gradient method for nonconvex optimization". In: *Open Journal of Mathematical Optimization* 5 (2024), pp. 1–22. DOI: 10.5802/ojmo.30. arXiv: 2208.00799.

[Sci+24b] Luca Sciullo, Alberto De Marchi, Angelo Trotta, Federico Montori, Luciano Bononi, and Marco Di Felice. "Relativistic Digital Twin: Bringing the IoT to the future". In: *Future Generation Computer Systems* 153 (2024), pp. 521–536. DOI: 10.1016/j.future.2023.12.016. arXiv: 2301.07390.

[BDR23] Andreas Britzelmeier, Alberto De Marchi, and Rebecca Richter. "Dynamic and nonlinear programming for trajectory planning". In: *IEEE Control Systems Letters* 7 (2023), pp. 2569–2574. ISSN: 2475-1456. DOI: 10.1109/LCSYS.2023.3285746.

[De 23a] Alberto De Marchi. "Proximal gradient methods beyond monotony". In: *Journal of Nonsmooth Analysis and Optimization* 4 (2023). DOI: 10.46298/jnsao-2023-10290. arXiv: 2211.04827.

[De 23b] Alberto De Marchi. "Regularized interior point methods for constrained optimization and control". In: *IFAC-PapersOnLine* 56.2 (2023). 22nd IFAC World Congress, pp. 1247–1252. DOI: 10.1016/j.ifacol.2023.10.1747. arXiv: 2210.15825.

[De +23a] Alberto De Marchi, Axel Dreves, Matthias Gerdts, Simon Gottschalk, and Sergejs Rogovs. "A function approximation approach for parametric optimization". In: *Journal of Optimization Theory and Applications* 196.1 (2023), pp. 56–77. DOI: 10.1007/s10957-022-02138-4.

[De +23b] Alberto De Marchi, Xiaoxi Jia, Christian Kanzow, and Patrick Mehlitz. "Constrained composite optimization and augmented Lagrangian methods". In: *Mathematical Programming* 201.1 (2023), pp. 863–896. DOI: 10.1007/s10107-022-01922-4. arXiv: 2203.05276.

[De 22] Alberto De Marchi. "On a primal-dual Newton proximal method for convex quadratic programs". In: *Computational Optimization and Applications* 81.2 (2022), pp. 369–395. DOI: 10.1007/s10589-021-00342-y.

[DT22] Alberto De Marchi and Andreas Themelis. "Proximal gradient algorithms under local Lipschitz gradient continuity". In: *Journal of Optimization Theory and Applications* 194.3 (2022), pp. 771–794. DOI: 10.1007/s10957-022-02048-5. arXiv: 2112.13000.

[Cal+21] Francesca Calà Campana, Alberto De Marchi, Alfio Borzí, and Matthias Gerdts. "On the numerical solution of a free end-time homicidal chauffeur game". In: *ESAIM: ProcS* 71 (2021), pp. 33–42. DOI: 10.1051/proc/202171104.

[De 19] Alberto De Marchi. "On the mixed-integer linear-quadratic optimal control with switching cost". In: *IEEE Control Systems Letters* 3.4 (2019), pp. 990–995. ISSN: 2475-1456. DOI: 10.1109/LCSYS.2019.2920425.

[DG19b] Alberto De Marchi and Matthias Gerdts. "Free finite horizon LQR: a bilevel perspective and its application to model predictive control". In: *Automatica* 100 (2019), pp. 299–311. ISSN: 0005-1098. DOI: 10.1016/j.automatica.2018.11.032.

[DG18] Alberto De Marchi and Matthias Gerdts. "Traffic flow on single-lane road networks: Multiscale modelling and simulation". In: *IFAC-PapersOnLine* 51.2 (2018). 9th Vienna International Conference on Mathematical Modelling (MATHMOD), pp. 162–167. DOI: 10.1016/j.ifacol.2018.03.028.

Conference proceedings

[De 26] Alberto De Marchi. "A condensing approach for linear-quadratic optimization with geometric constraints". In: *2026 European Control Conference (ECC)*. 2026, pp. 3833–3838. arXiv: 2510.17465.

[DO26] Alberto De Marchi and Dominique Orban. *Envelopt: Constrained convex composite optimization*. Tech. rep. G-2026-42. GERAD, 2026. arXiv: 2609.29521.

[Cia+25] Leonardo Ciabattini, Luca Sciullo, Alberto De Marchi, Lorenzo Gigli, Luciano Bononi, and Marco Di Felice. "Generative digital twin for predictive modeling in dynamic IoT scenarios". In: *2025 IEEE International Conference on Smart Computing (SMARTCOMP)*. Cork, Ireland, 2025, pp. 98–105. DOI: 10.1109/SMARTCOMP65954.2025.00071.

[RDG24] Rebecca Richter, Alberto De Marchi, and Matthias Gerdts. "Collision avoidance using iterative dynamic and nonlinear programming with adaptive grid refinements". In: *2024 European Control Conference (ECC)*. 2024, pp. 52–57. DOI: 10.23919/ECC64448.2024.10591064. arXiv: 2311.03148.

[Sci+24a] Luca Sciullo, Alberto De Marchi, Lorenzo Gigli, Monica Palmirani, and Fabio Vitali. "AAA: A blockchain-based architecture for ethical, robust authenticated anonymity". In: *Proceedings of the 2024 International Conference on Information Technology for Social Good*. GoodIT '24. Bremen, Germany: Association for Computing Machinery, 2024, pp. 1–9. DOI: 10.1145/3677525.3678676.

[De 21b] Alberto De Marchi. "Augmented Lagrangian methods as dynamical systems for constrained optimization". In: *2021 60th IEEE Conference on Decision and Control (CDC)*. 2021. DOI: 10.1109/CDC45484.2021.9683199.

[De 20] Alberto De Marchi. "Constrained and sparse switching times optimization via augmented Lagrangian proximal methods". In: *2020 American Control Conference (ACC)*. Denver, CO, USA: IEEE, 2020, pp. 3633–3638. DOI: 10.23919/ACC45564.2020.9147892.

[DG20] Alberto De Marchi and Matthias Gerdts. "Sparse switching times optimization and a sweeping Hessian proximal method". In: *Operations Research Proceedings 2019*. Ed. by Janis S. Neufeld, Udo Buscher, Rainer Lasch, Dominik Möst, and Jörn Schönberger. Cham: Springer International Publishing, 2020, pp. 89–95. ISBN: 978-3-030-48439-2. DOI: 10.1007/978-3-030-48439-2_11.

[DG19a] Alberto De Marchi and Matthias Gerdts. "An iterative method for final time optimization in nonlinear optimal control". In: *2019 Proceedings of the Conference on Control and its Applications*. SIAM, 2019, pp. 60–66. DOI: 10.1137/1.9781611975758.10.

[DG19c] Alberto De Marchi and Matthias Gerdts. "Nonsmooth Newton's method: Some structure exploitation". In: *ICCS 2019, LNCS 11538*. Ed. by J. M. F. Rodrigues et al. Springer, 2019, pp. 409–420. DOI: 10.1007/978-3-030-22744-9_32.

[BDG18] Andreas Britzelmeier, Alberto De Marchi, and Matthias Gerdts. "An iterative solution approach for a bi-level optimization problem for congestion avoidance on road networks". In: *Numerical Methods for Optimal Control Problems*. Ed. by Maurizio Falcone, Roberto Ferretti, Lars Grüne, and William M. McEneaney. Springer International Publishing, 2018, pp. 23–38. DOI: 10.1007/978-3-030-01959-4_2.

[Bir+16] Francesco Biral, Enrico Bertolazzi, Paolo Bosetti, Alberto De Marchi, and Martin M. Hanczyc. "Optimal control of a laser source to generate a minimum time trajectory of a droplet in a liquid layer". In: *2016 Future Technologies Conference (FTC)*. 2016, pp. 1208–1216. DOI: 10.1109/FTC.2016.7821754.

Theses

[De 21a] Alberto De Marchi. "Augmented Lagrangian and Proximal Methods for Constrained Structured Optimization". PhD thesis. Munich, Germany: University of the Bundeswehr Munich, 2021. DOI: 10.5281/zenodo.4972535.

[De 16] Alberto De Marchi. "Modelling and optimal control of laser-driven thermocapillary motion of a droplet". MA thesis. University of Trento, 2016. DOI: 10.5281/zenodo.1481663.

Preprints

[AD26] Behzad Azmi and Alberto De Marchi. *Interior-point proximal methods for nonsmooth optimization in Hilbert spaces with cone-ordered constraints*. 2026. arXiv: 2609.13813.

[BDS26] Enrico Bertolazzi, Alberto De Marchi, and Davide Stocco. *Dynamic proximal point method for unconstrained minimization*. 2026. arXiv: 2608.03349.

[BDM26] Ernesto G. Birgin, Alberto De Marchi, and Patrick Mehlitz. *Elastically safeguarded augmented Lagrangian methods*. 2026. arXiv: 2607.02697.

[Cal+26] Andrew Calcan, Jordan Collard, Alberto De Marchi, and Scott B. Lindstrom. *Resolvent Moreau identities without monotonicity: theory and applications to Gabay duality, Douglas–Rachford and ADMM*. 2026. arXiv: 2605.18158.

[DDK26] Simeon vom Dahl, Alberto De Marchi, and Christian Kanzow. *Proximal limited-memory quasi-Newton methods for nonsmooth nonconvex optimization*. 2026. arXiv: 2605.11627.

[DMT26] Alberto De Marchi, Yura Malitsky, and Adrien B. Taylor. *Mirror descent algorithms with logarithmic barriers*. 2026. arXiv: 2608.22834.

[DO26] Alberto De Marchi and Dominique Orban. *Envelopt: Constrained convex composite optimization*. Tech. rep. G-2026-42. GERAD, 2026. arXiv: 2609.29521.

[Mén+26] Etienne Ménager, Pierre Fabre, Antoine Bambade, Wilson Jallet, Alberto De Marchi, and Justin Carpentier. *Frictional Contact-Implicit Inverse Dynamics*. 2026. HAL: hal-05201780.

[BD25] Hòa T. Bùi and Alberto De Marchi. *Cutting plane methods with gradient-based heuristics*. 2025. arXiv: 2511.00520.

[De 24a] Alberto De Marchi. *Affordable mixed-integer Lagrangian methods: optimality conditions and convergence analysis*. 2024. arXiv: 2406.12436.